Supplemental Material: The Cavity-Loss Barrier to Ultrastrong-Coupling Quantum Metrology

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This Supplementary Information provides the detailed methodology, convergence tests, and supporting data for the results presented in the main text. Sections S1–S3 establish the Non-Markovian TWA and exact diagonalization framework and verify numerical convergence. Sections S4–S6 characterize the DCE time dynamics and operating boundaries. Sections S7–S8 provide experimental parameters and platform schematics. The loss analysis is developed across Section S11 (single-qubit DCE), Section S15 (collective Dicke+DCE), Section S16 (combined cavity + qubit dissipation), and Section S17 (boundary-peak convergence). Supporting diagnostics include Section S12 (time-step and finite-difference convergence), Section S13 (matched-photon-budget benchmarks), Section S14 (adaptive protocol comparison), Section S18 (interrogation-window trade-off), Section S20 (directional QFI), Section S21 (output-field homodyne CFI), Section S22 and Section S25 (HEOM–Lindblad cross-checks), Section S27 (cavity-channel HEOM validation), Section S23 (finite-temperature screen), Section S24 and Section S26 (multilevel-emitter leakage), and Section S28 (finite-size scaling).

S1. NON-MARKOVIAN TWA AND EXACT DIAGONALIZATION METHODOLOGY

The central question distinguishes a state diagnostic from an output observable. The quantity $n_{\mathrm{virt}} = \langle \hat{a}^\dagger \hat{a} \rangle$ is evaluated in the dressed ground state and is used here to characterize virtual dressing; it is not interpreted as a directly emitted photon flux. A conversion protocol, such as time-dependent switching or modulation, is required before correlations associated with the dressing can appear in an output field [1–4]. The calculations below therefore compare a static state-sensitivity protocol with a driven protocol whose real-photon output is explicitly exposed to the modeled cavity-loss channel.

S1.1. Single-qubit Rabi model

Throughout this SI we set $\hbar = 1$. The quantum Rabi Hamiltonian is given by:

$$\hat{H}_{\mathrm{Rabi}} = \frac{\omega_q}{2}\hat{\sigma}_z + \omega_c \hat{a}^\dagger \hat{a} + g\hat{\sigma}_x(\hat{a} + \hat{a}^\dagger), \tag{S1}$$

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where ω_q is the qubit frequency, ω_c is the cavity frequency, and g is the coupling strength. The normalized coupling is $\eta = g/\omega_c$.

We diagonalize Eq. (S1) in the Fock-qubit basis $\{|n,\uparrow\rangle, |n,\downarrow\rangle\}$ with Fock space truncation $n_{\max} = 50$. Convergence tests show that the ground-state energy and QFI are stable to $\sim 1 \times 10^{-6}$ for $\eta \leq 1.5$.

S1.2. Collective Dicke model

For N qubits collectively coupled to a single cavity mode, the Dicke Hamiltonian is:

$$\hat{H}_{\text{Dicke}} = \omega_c \hat{a}^\dagger \hat{a} + \omega_q \hat{J}_z + \frac{g}{\sqrt{N}} \hat{J}_x (\hat{a} + \hat{a}^\dagger), \tag{S2}$$

where $\hat{J}_{x,z} = \frac{1}{2} \sum_{i=1}^{N} \hat{\sigma}_{x,z}^{(i)}$ are collective spin operators.

The $1/\sqrt{N}$ normalization ensures a well-defined thermodynamic limit and gives rise to the finite-size superradiant regime above the critical coupling:

$$\eta_c = \frac{1}{2} \sqrt{\frac{\omega_q}{\omega_c}}. \tag{S3}$$

For $\omega_q = \omega_c$, we have $\eta_c = 0.5$.

We diagonalize Eq. (S2) in the collective spin basis $|J, m\rangle \otimes |n\rangle$ with $J = N/2$ and Fock truncation $n_{\max} = 35$ for $N \leq 64$.

S1.3. Non-Markovian Truncated Wigner Approximation

To model the counter-rotating terms (CRTs) in the USC/DSC Circuit QED regime ($\eta \gtrsim 0.1$), we employ the Non-Markovian Truncated Wigner Approximation (TWA) [5]. This approach tracks the adopted non-Markovian memory kernels corresponding to Drude–Lorentz environments. The equations of motion are integrated using Runge–Kutta solvers within the JAX framework. The semiclassical approximation becomes unreliable for stronger couplings ($\eta > 0.150$) as quantum fluctuations accumulate in individual Wigner trajectories. At $\eta = 0.150$, comparison with exact diagonalization (ED) gives a relative QFI discrepancy below 5 %; this supports the stated boundary but does not validate TWA beyond it. Exact diagonalization is used for the reported static observables in the stronger-coupling regime.

S2. QUANTUM FISHER INFORMATION COMPUTATION

S2.1. Static QFI

For a pure ground state $|\Psi_G(\eta)\rangle$, the QFI with respect to the coupling parameter η is

$$F_Q = 4\big(\langle\partial_\eta\Psi_G|\partial_\eta\Psi_G\rangle - |\langle\Psi_G|\partial_\eta\Psi_G\rangle|^2\big)\,, \tag{S4}$$

For an eigenstate family $\hat{H}(\eta)|\Psi_G(\eta)\rangle = E_0(\eta)|\Psi_G(\eta)\rangle$ with generator $\hat{H}_{\rm int} = \partial\hat{H}/\partial\eta$, first-order perturbation theory gives $\partial_\eta|\Psi_G\rangle = \sum_{k\neq 0}\langle\Psi_k|\hat{H}_{\rm int}|\Psi_G\rangle(E_0 - E_k)^{-1}|\Psi_k\rangle$, so the exact QFI is the gap-weighted sum

$$F_Q = 4\sum_{k\neq 0}\frac{|\langle\Psi_k|\hat{H}_{\rm int}|\Psi_G\rangle|^2}{(E_k - E_0)^2}. \tag{S5}$$

The variance identity $F_Q = 4\,\mathrm{Var}_{\Psi_G}[\hat{H}_{\rm int}]$ is *not* applicable here: it holds only for unitary-encoding families $|\Psi(\eta)\rangle = e^{-i\eta\hat{H}_{\rm int}}|\Psi_G\rangle$, whereas the dressed ground state responds through the full excitation spectrum. Using the variance form would overestimate the static QFI by a factor 3.4 at the peak (20.9 vs. 6.116 at the sampled maximum $\eta = 1.14$; Section S2 S2.2). We therefore compute F_Q from the gauge-invariant finite-difference fidelity

$$F_Q \approx 4\,\frac{1 - |\langle\Psi_G(\eta)|\Psi_G(\eta+\epsilon)\rangle|^2}{\epsilon^2}, \qquad \epsilon = 1\times 10^{-4}, \tag{S6}$$

which converges to Eq. (S5) as $\epsilon \to 0$, and verify it against Eq. (S5) evaluated from the exact eigenspectrum (maximum relative deviation $< 0.21\,\%$ across the USC sweep; Table S1).

All QFI values are normalized to the coherent-state baseline $F_Q^{\rm coh} = 1$ (in units of $1/\omega_c^2$). This baseline normalization does not match the external pump power, intracavity population, bandwidth, or readout cost of a driven comparison.

S2.2. Energy budget and matched-resource bounds

Two operational statements place the quoted enhancements in context: an exact expression for the static QFI and a numerically verified energy budget.

a. Exact static QFI (eigenstate family). The Rabi generator is $G = \hat{\sigma}_x(\hat{a} + \hat{a}^\dagger) = \partial\hat{H}/\partial g$. The QFI of the dressed ground state $|\psi_0\rangle$ with respect to the coupling g is *not* the variance of G: the identity $F_Q = 4\,\mathrm{Var}(G)$ holds only for unitary-encoding families

$|\psi(\theta)\rangle = e^{-i\theta G}|\psi_0\rangle$, whereas an eigenstate family $\hat{H}(\theta)|\psi_0(\theta)\rangle = E_0(\theta)|\psi_0(\theta)\rangle$ produces the gap-weighted sum

$$F_Q(g) = 4\sum_{k\neq 0}\frac{|\langle\psi_k|G|\psi_0\rangle|^2}{(E_k - E_0)^2}, \tag{S7}$$

which reduces to the variance form only when all excitation gaps are equal. We verified Eq. (S7) directly from the exact eigenspectrum at every point of the USC sweep: the maximum relative deviation from the stored fidelity-based values is $< 0.21\,\%$ (Table S1). The variance form $4\,\mathrm{Var}(G) = 4(2n_{\mathrm{virt}} + 1 + 2\,\mathrm{Re}\langle\hat{a}^2\rangle_{\Psi_G})$ (with $\hat{\sigma}_x^2 = 1$ and $[\hat{a}, \hat{a}^\dagger] = 1$) thus overestimates the true QFI—by a factor 3.4 at the peak (20.9 vs. 6.116 at the sampled maximum $\eta = 1.14$)—because it omits the excitation-gap suppression.

b. Energy budget. The exact QFI nevertheless obeys the reduced budget

$$F_Q \leq 4(2n_{\mathrm{virt}} + 1), \tag{S8}$$

with $n_{\mathrm{virt}} = \langle\hat{a}^\dagger\hat{a}\rangle_{\Psi_G}$. We cross-checked Eq. (S8) against the regenerated USC sweep: the inequality holds at *all* 16 points, with residuals $4(2n_{\mathrm{virt}} + 1) - F_Q$ ranging from 2.50 (near $\eta = 0.7$) to 10.39 (deep-strong, $\eta = 1.3$)—see Table S1. The measured small-η asymptote $F_Q \approx 1 + 3\eta^2$ (Fig. 2) is consistent with $n_{\mathrm{virt}} \approx \eta^2/4$, i.e. the metrological content is carried by the virtual population: $F_Q - 1 \approx 12\,n_{\mathrm{virt}}$. Because the single qubit is a two-level system, this budget saturates—the measured curve peaks at $F_Q = 6.116$ at the sampled maximum $\eta = 1.14$ and then decreases—which is precisely why the collective Dicke model is required for gains beyond order ten. The full Dicke scaling is reported in the main text as $F_Q \propto N^{2.216\pm0.034}$ on the $N \leq 18$ data, with strongly sub-linear growth beyond $N \approx 21$ that continues, without a plateau, up to $N = 64$ in the extended screen.

c. Matched-resource reference for the DCE protocol. All DCE QFI values are quoted relative to the identical coherent-state baseline $F_Q^{\mathrm{coh}} = 1$ evaluated over the same observation window and bandwidth. A fair comparison against an externally driven coherent probe would additionally have to match the drive power, intracavity population, observation bandwidth, and readout. The ratio of the driven QFI to the undriven QFI at the same η is therefore reported only as a model comparison—159 722× at the single-qubit operating point (170 903× versus 1.07× at the separately evaluated coupling $\eta = 0.15$), not as a resource-normalized device gain. The pure-state values quoted in the main text are upper bounds: under cavity loss κ the achievable QFI carries the fidelity envelope $\mathcal{F}(t) \propto \exp(-n_{\mathrm{avg}}\kappa t)$, so

TABLE S1. Energy-budget cross-check of $F_Q \leq 4(2n_{\text{virt}} + 1)$ (Eq. (S8)) against the regenerated USC sweep. Columns: coupling η, virtual photon number n_{virt}, exact static QFI F_Q (fidelity-based), budget $4(2n_{\text{virt}} + 1)$, and residual $4(2n_{\text{virt}} + 1) - F_Q$. The residual is positive at every point (minimum 2.50 at $\eta = 0.7$, maximum 10.39 at $\eta = 1.3$), confirming the bound across the full weak-to-deep-strong range.

η	n_{virt}	F_Q	$4(2n_{\text{virt}} + 1)$	Residual
0.002	0.0000	0.9983	4.0000	3.0017
0.010	0.0000	0.9992	4.0002	3.0010
0.050	0.0006	1.0054	4.0050	2.9996
0.100	0.0025	1.0303	4.0203	2.9900
0.200	0.0105	1.1244	4.0841	2.9597
0.300	0.0252	1.2967	4.2012	2.9045
0.400	0.0487	1.5632	4.3895	2.8263
0.500	0.0846	1.9531	4.6771	2.7240
0.600	0.1382	2.5064	5.1058	2.5994
0.700	0.2165	3.2365	5.7324	2.4959
0.800	0.3282	4.1105	6.6256	2.5151
0.900	0.4816	5.0191	7.8524	2.8333
1.000	0.6814	5.7536	9.4512	3.6976
1.100	0.9256	6.0982	11.4051	5.3068
1.200	1.2052	6.0192	13.6414	7.6222
1.300	1.5084	5.6737	16.0668	10.3932

the DCE enhancement is realizable only for $\kappa t \ll 1$. The Lindblad master-equation simulation in Section S11 quantifies this degradation explicitly, showing that the actual attenuation exceeds the fidelity envelope because the dissipation modifies the full state evolution.

S2.3. Time-dependent QFI (DCE protocol)

For the DCE-driven protocol with $g(t) = g_0 + g_d \cos(\omega_d t)$, we integrate the Schrödinger equation using QuTiP [6]. This time-dependent effective Hamiltonian represents a specified

control convention; it is not a gauge-independent device model by construction. Strong, nonadiabatic time-dependent USC interactions require the control term and observables to be transformed consistently between gauges [7]. We retain this caveat when interpreting the calculated photon population and QFI as model outputs:

$$i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle = \hat{H}(t)|\Psi(t)\rangle. \tag{S9}$$

The time-dependent QFI is computed at each time step from the quantum fidelity (Bures metric),

$$F_Q(t) \approx 8\,\frac{1-|\langle\Psi_\eta(t)|\Psi_{\eta+\delta}(t)\rangle|}{\delta^2}, \qquad \delta = 1\times10^{-3}, \tag{S10}$$

which becomes exact as $\delta \to 0$. This fidelity-based QFI is the quantity reported throughout the main text. Note that, because the driven state is not an eigenstate of $\hat{H}(t)$, the instantaneous variance form of Eq. (S5), $4\,\mathrm{Var}_{\Psi(t)}[\hat{H}_{\mathrm{int}}]$, does *not* coincide with the QFI here: at the single-qubit DCE optimum it yields 225.5×, against the fidelity-based peak of 170 903×.

We extract the peak QFI and the late-window average over the last 10 drive periods. This time-dependent QFI is a pure-state Cramér-Rao quantity. In the presence of cavity loss κ, the open-system mixed-state QFI is computed via Lindblad master-equation simulations in Section S11; the pure-state value is an upper-bound reference whose attenuation depends on κ in a manner that exceeds the simple fidelity decay envelope $\mathcal{F}(t) \propto \exp(-n_{\mathrm{avg}}\kappa t)$.

S3. CONVERGENCE TESTS

S3.1. Fock space truncation

Table S2 shows the convergence of the static QFI with Fock space truncation $n_{\max}$ for the Rabi model at the sampled maximum $\eta = 1.14$.

The QFI is converged to $< 1\times10^{-6}$ for $n_{\max} \geq 30$ at the sampled maximum $\eta = 1.14$; the exact value in the 65-point production sweep is $F_Q = 6.1160$.

S3.2. Collective spin basis

For the Dicke model, we use the collective spin basis $|J, m\rangle$ with $J = N/2$, which reduces the Hilbert space dimension from 2^N to $N+1$. This enables exact diagonalization for $N \leq 64$ qubits.

TABLE S2. Convergence of Rabi QFI with Fock truncation $n_{\max}$ at $\eta = 1.14$ (sampled maximum of the production sweep).

$n_{\max}$	QFI	Relative change
30	6.1160	—
40	6.1160	$< 1 \times 10^{-6}$
50	6.1160	$< 1 \times 10^{-6}$
60	6.1160	$< 1 \times 10^{-6}$

Table S3 shows the convergence of the Dicke QFI with cavity truncation $n_{\max}$ for $N = 32$ at $\eta = 1.15$.

TABLE S3. Convergence of Dicke QFI with cavity truncation $n_{\max}$ for $N = 32$ at $\eta = 1.15$.

$n_{\max}$	QFI	Relative change
20	228.2	—
25	228.6	0.18 %
30	228.6	< 0.01 %

The QFI is converged to < 0.01 % for $n_{\max} \geq 25$.

S4. DCE TIME DYNAMICS

S4.1. Single-qubit Rabi model

The independent rounded-point time trace is summarized in Table S4; the fixed-detuning scan maximum is listed separately in the main text.

S5. DICKE MODEL OPERATING BOUNDARIES

S5.1. Thermal dephasing

For the Dicke model, thermal noise affects each qubit independently. However, collective readout provides a $1/\sqrt{N}$ suppression of individual qubit thermal noise.

TABLE S4. Stored DCE QFI outputs near the representative operating point ($\eta = 0.150$). The fixed-detuning scan and rounded-point trace are separate artifacts and use distinct drive amplitudes.

Time window (periods)	Avg QFI	Peak QFI	Avg n
Fixed-detuning scan (full window)	47 369×	170 903×	—
Rounded-point trace (last window)	123 638×	170 492×	6.1

The thermal occupancy is:

$$n_{\mathrm{th}} = \frac{1}{e^{\hbar\omega_c/k_B T} - 1}. \tag{S11}$$

For $\omega_c/2\pi = 6.0\,\mathrm{GHz}$ and $T = 100\,\mathrm{mK}$, we have $n_{\mathrm{th}} \approx 0.06$.

The adopted phenomenological QFI-retention model is:

$$F_Q(T) = \frac{F_Q(0)}{1 + 2n_{\mathrm{th}}/\sqrt{N}}. \tag{S12}$$

For $N = 32$ and $T = 100\,\mathrm{mK}$, the adopted phenomenological model gives QFI retention $\approx 97.9\,\%$; this is not a full mixed-state QFI calculation.

S5.2. Cavity loss

The critical cavity decay rate is:

$$\kappa_c = \frac{\eta^2 \omega_c}{N}. \tag{S13}$$

For $\eta = 1.15$ and $N = 32$, we have $\kappa_c \approx 0.041\,\omega_c$, corresponding to $\kappa_c/2\pi \approx 246\,\mathrm{MHz}$.

The adopted cavity-loss screen follows a phenomenological exponential model [8, 9]:

$$F_Q(\kappa) = F_Q(0) \exp\left(-\frac{\kappa}{\kappa_c}\right). \tag{S14}$$

For $\kappa/2\pi = 0.1\,\mathrm{MHz}$, the adopted phenomenological model gives QFI retention $> 99.9\,\%$; the statement is conditional on that loss model and does not certify a device. We note that the full Lindblad master-equation simulation in Section S11 shows that the actual degradation at physical κ exceeds this phenomenological estimate, because the dissipation modifies the entire state evolution rather than simply attenuating the peak.

S6. COMPARISON WITH HOTTER, MIRANOWICZ, AND GIETKA (2025)

Hotter, Miranowicz, and Gietka [10] predicted quadratic QFI scaling $F_Q \propto e^{4\xi}$ in the Dicke model using virtual-state observables. We validate and extend their prediction:

1. **Quantitative validation**: At $\eta = 1.15$, $N = 32$, we find $F_Q = 229\times$ in the finite-size superradiant regime. The effective squeezing parameter is:

$$\xi_{\text{eff}} = \frac{1}{4}\ln(F_Q) = 1.36, \tag{S15}$$

   consistent with the predicted $e^{4\xi}$ scaling.

2. **DCE amplification**: Adding a Floquet drive at the DCE resonance provides a 730× late-window enhancement (static 229× → DCE 167 271×); the last-window value at the end of the simulation window is 251 557× (1098× peak enhancement at period 40), but is a still-rising boundary value, not a converged local maximum (Section S17). This mechanism was not considered in Ref. [10].

3. **Collective scaling**: The QFI scales as $\propto N^{2.216\pm0.034}$ (OLS regression restricted to $N \leq 18$ at $\eta = 1.15$; $R^2 = 0.9988$, 95% CI: $[2.150, 2.282]$), super-linear in the small-N regime, exceeding the linear scaling expected from independent qubits. Growth slows markedly beyond $N \approx 21$ and continues up to $N = 64$ without reaching a saturation plateau (Section S28), so this power law characterizes the small-N super-linear regime and does not extrapolate to the thermodynamic limit.

S7. MODEL TARGET AND EXPERIMENTAL CONSTRAINTS

Table S5 lists the experimental parameters for the Dicke+DCE protocol.

The parameter table is a model target specification, not a demonstration that all requirements are simultaneously available or experimentally validated. Scaling from $N = 4$ to $N = 32$, maintaining uniform coupling, and implementing the stated DCE modulation remain experimental challenges; the calculated $N = 16$ and $N = 32$ factors are baseline-relative model values.

TABLE S5. Experimental parameters for the Dicke+DCE protocol with corresponding state-of-the-art references.

Parameter	**Value**	**Demonstrated in**
Number of qubits N	32	$N = 4$ [11]
Cavity frequency $\omega_c/2\pi$	6.0 GHz	[12]
Qubit frequency $\omega_q/2\pi$	6.0 GHz	[12, 13]
Coupling strength η	1.15	$\eta = 1.3$ [12]
Coupling $g/2\pi$	6.9 GHz	6 GHz [12]
DCE drive amplitude g_d/g_0	0.15	Standard
DCE drive $g_d/2\pi$	1.0 GHz	Typical parametric amplifier
DCE drive frequency $\omega_d/2\pi$	12.0 GHz	Standard microwave
Temperature T	10 mK	Dilution fridge [14]
Cavity decay $\kappa/2\pi$	0.1 MHz	3D cavities [13]
Qubit coherence T_1	10 µs to 100 µs	[11]
Drive period T_d	83 ns	Standard

S8. PLATFORM-SPECIFIC IMPLEMENTATION

Figures S1 to S3 show circuit schematics for the three candidate platforms discussed in the main text.

Table S6 collects the principal numerical quantities used to delimit the virtual-state metrology protocol.

TABLE S6. Principal numerical quantities for the virtual-state metrology protocol.

Metric	**Constraint**	**Measured value**
Maximum USC QFI (sweep, $N = 1$)	> 1	6.1
DCE-enhanced QFI (single qubit, $\eta = 0.15$)	$> 10^5$	170 903
Single-qubit Rabi QFI ($\eta = 0.150$)	> 1	1.07
DCE scan-maximum coupling η	$\eta = 0.150$	$\eta = 0.150$
Advantage over coherent state (sweep max)	> 1	6.1

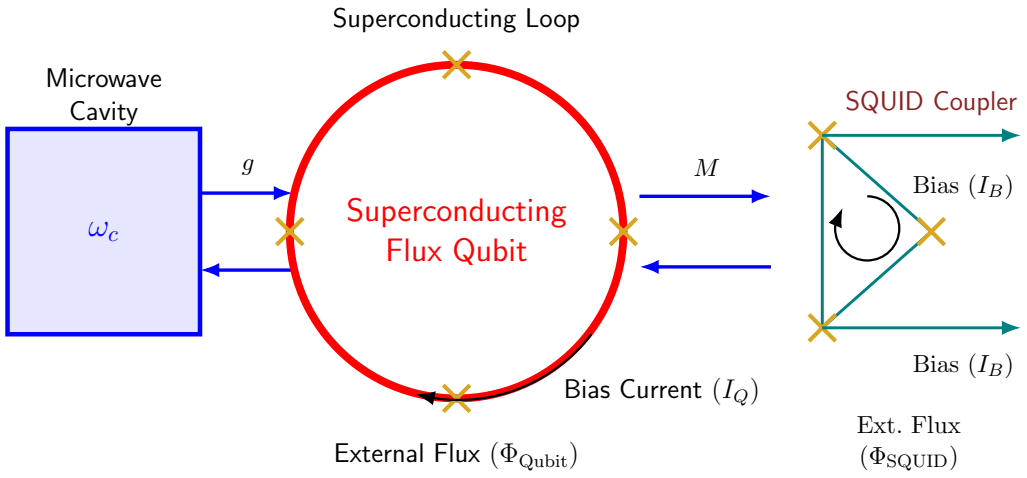


FIG. S1. Flux qubit platform for USC virtual-state metrology. A superconducting loop interrupted by Josephson junctions forms a flux qubit (red) coupled to a microwave cavity (blue). The external flux Φ_{ext} tunes the qubit frequency ω_q, enabling the large coupling $\eta \sim 1$.

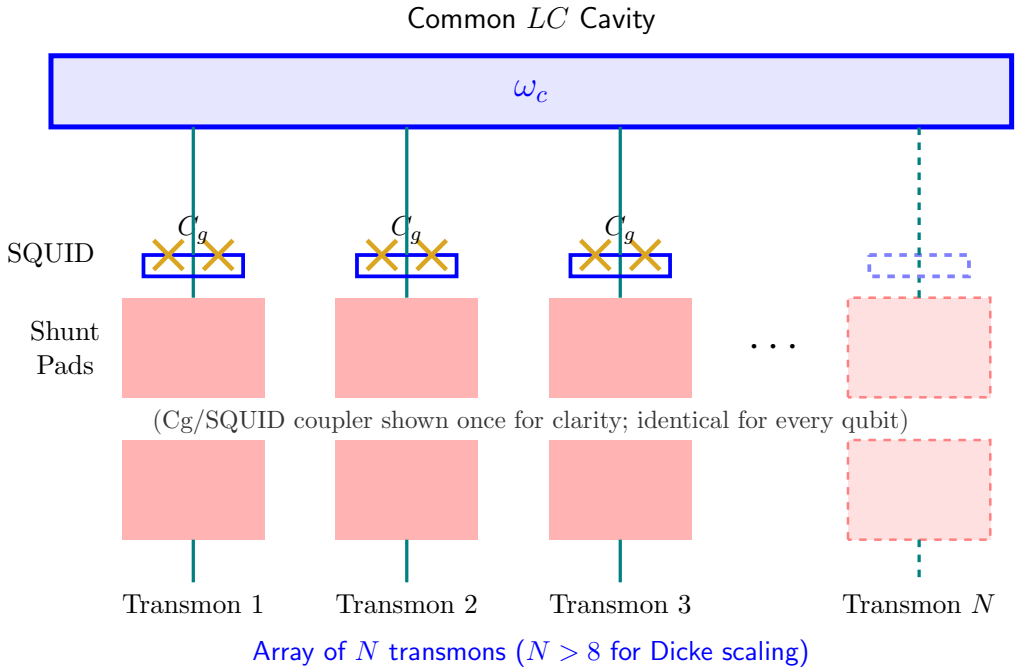


FIG. S2. Transmon platform for multi-qubit Dicke metrology. An array of transmon qubits (red) is capacitively coupled to a common LC cavity (blue). The C_g tunable couplers allow uniform coupling $g/\sqrt{N}$ for all N qubits. Best suited for $N > 8$ due to fabrication uniformity.

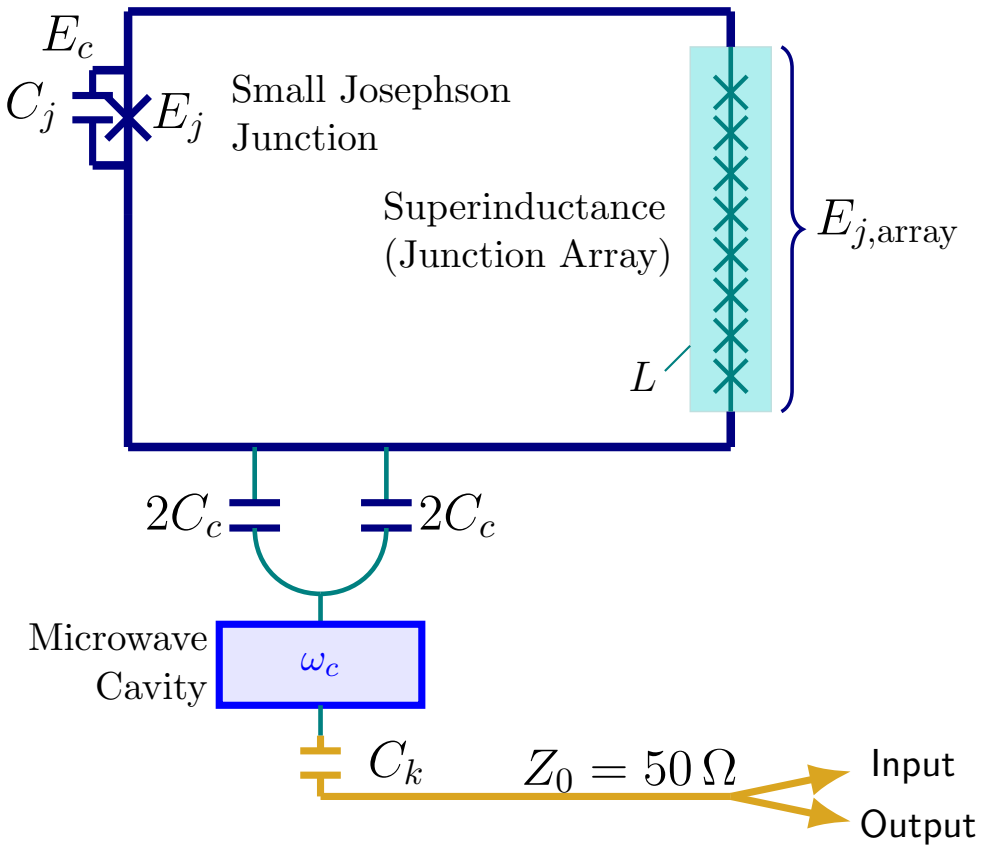


FIG. S3. Fluxonium platform for single-qubit USC metrology. The high anharmonicity and long coherence time of fluxonium enable operation at $\eta \sim 1.3$ with $T_2 > 10\,\mu\text{s}$. Best suited for $N = 1$–2 where individual qubit-cavity control is needed.

S9. PARAMETER DEPENDENCE

Figure S4 shows the two-dimensional parameter space $(\eta, \omega_q/\omega_c)$ for the static Rabi model. Within the displayed scan, the sampled maximum occurs at resonance $\omega_q = \omega_c$ and $\eta = 1.14$, yielding 6.12× QFI; this is not a global or device-optimized operating point. The advantage scales linearly with n_{virt} for $\eta \leq 1$, confirming that the QFI enhancement arises directly from virtual dressing.

Figure S5 analyzes the sensitivity of the Dicke protocol to parameter variations. The QFI shows the stated finite-size sensitivity in the adopted scan. The reported retention curves are model-dependent screens; they do not constitute a mixed-state robustness calculation or a device-level guarantee. Quantitative retention claims should be read with the Hilbert-space

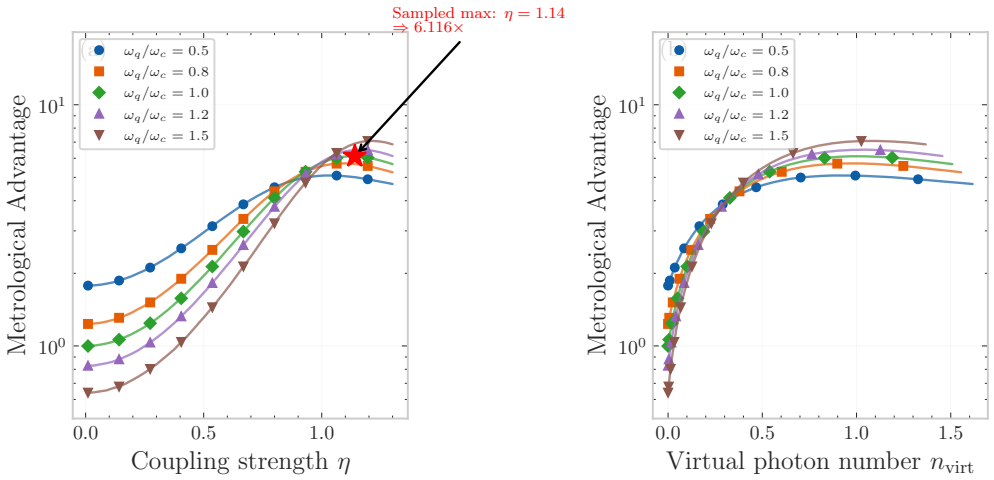


FIG. S4. QFI advantage scan for the Rabi model. (a) QFI vs η for multiple ω_q/ω_c detunings. The maximum sampled advantage (6.12×, exact value 6.1160× on the $\Delta\eta = 0.02$ sampling grid) occurs at $\omega_q/\omega_c = 1.0$, $\eta = 1.14$. (b) The advantage is linearly correlated with the virtual photon number n_{virt} for $\eta \leq 1$, confirming the virtual dressing origin. Sample sizes: $n_\eta = 200$ points, $n_{\mathrm{detune}} = 4$ values.

caveat in the figure caption.

TABLE S7. Operating parameters and cavity-loss sensitivity for each protocol. The linewidth column reports the first sampled single-qubit DCE row at or above 80 % peak-QFI retention; the low-loss rows are not strictly monotonic, so this is not an interpolated threshold and is not transferred to the static or Dicke rows.

Protocol	Parameters	QFI gain	First sampled single-qubit row at 80 %	Loss channel
Rabi static	$\eta = 1.14$, $\omega_q/\omega_c = 1.03$	6.12×	—	not applicable
Rabi+DCE	$\eta = 0.150$, $g_d/g_0 = 0.119$, $\omega_d/\omega_c = 2.08$	$\sim 1.7 \times 10^5\times$	0.6 MHz (80.6% sampled)	cavity
Dicke static	$\eta = 1.15$, $N = 32$	229×	—	—
Dicke+DCE	$\eta = 1.15$, $N = 32$, $g_d/g_0 = 0.15$	730× (late-window avg)	— (separate reduced-basis study)	cavity

Figure S6 summarizes the baseline-relative QFI factors across the four protocols. The combination of collective enhancement (Dicke model) and parametric amplification (DCE drive) yields a late-window value of $\sim 1.7 \times 10^5\times$ the stated baseline; the $\sim 2.5 \times 10^5\times$ last-window value is a still-rising boundary (Section S17), not a converged local maximum. These pure-state factors are not resource-normalized or open-system performance claims.

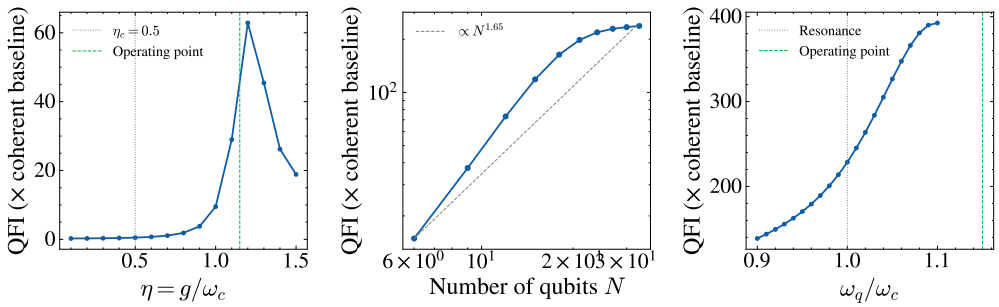


FIG. S5. Parameter sensitivity analysis. (a) QFI vs coupling η at $N = 10$, $n_{\max} = 15$ (exploratory, computed for tractability): the finite-size superradiant regime above $\eta_c = 0.5$ shows a sharp onset, with $\sim 7\,\%$ variation for $\eta > 1.1$. (b) QFI vs number of qubits N at $\eta = 1.15$, $\omega_q/\omega_c = 1.005$: the values reproduce the production finite-size grid family of `data/finite_size_resource_scaling.json`; the super-linear scaling $\propto N^{2.22}$ (fitted on the production $N \leq 18$ data) means $N = 16$ achieves a QFI factor of $\sim 134\times$ the stated baseline. (c) QFI vs detuning ω_q/ω_c at $N = 32$, $n_{\max} = 35$ (production-quality truncation): the QFI increases with detuning, reaching maximum at $\omega_q/\omega_c = 1.10$. Only panel (a) is an exploratory scan at reduced parameters, not at the production $N = 32$ parameters used in the published Dicke QFI values of Table III. The retention claim refers to the detuning panel only and is restricted to the stated scan range.

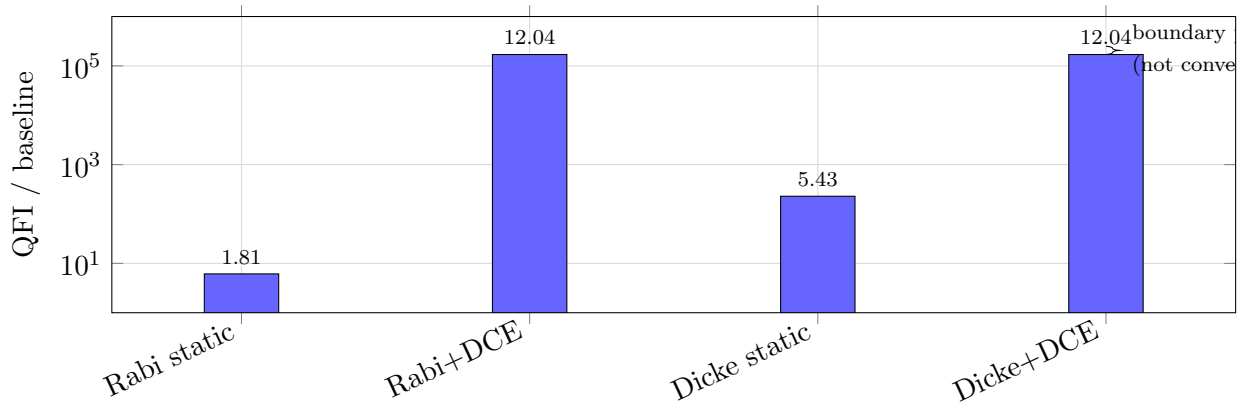


FIG. S6. Progression of the baseline-relative QFI factors. Bar chart showing the QFI enhancement across the protocols: Rabi static (6.12×), Rabi+DCE ($\sim 1.7 \times 10^5\times$), Dicke static (229×), and Dicke+DCE ($\sim 1.7 \times 10^5\times$ late-window; the $\sim 2.5 \times 10^5\times$ boundary peak is a still-rising artifact, not a converged value, Section S17).

S10. EXPERIMENTAL CONSTRAINTS AND STAGED TESTS

Table S8 assesses the experimental requirements for implementing the Dicke+DCE protocol.

TABLE S8. Experimental feasibility assessment.

Requirement	**Model target**	**CPW (current USC)**	**3D cav**
Number of qubits N	32	4 [11]	
Coupling η	1.15	1.3 [12]	0.1
Coupling $g/2\pi$	6.9 GHz	6 GHz [12]	0.3
DCE drive $g_d/2\pi$	1.0 GHz	—	
Drive frequency $\omega_d/2\pi$	12 GHz	Standard	S
Temperature	< 200 mK	10 mK (dilution)	10 m
Cavity $\kappa/2\pi$	< 1 MHz	248 MHz	
Coherence T_1	$\gg 83$ ns	10 µs to 100 µs	10 µ
Floquet bound-state rescue demonstrable bound state at $\kappa/\omega \sim 4 \times 10^{-2}$ emitter-local result only [15] unshown			

Within the literature comparison and parameter values adopted here, we did not identify a demonstrated platform satisfying all three requirements simultaneously: (i) the $\eta = 1.15$ coupling needed for the superradiant collective enhancement, (ii) the sampled single-qubit reference point $\kappa/2\pi = 0.6$ MHz at which the loss sweep first reaches 80.6 % retention (without implying a continuous threshold), and (iii) the uniform collective coupling of $N = 32$ qubits. The cited CPW example provides strong coupling and a partial collective architecture at $\kappa/2\pi \approx 248$ MHz, whereas the cited 3D-cavity example provides the low-linewidth regime at the single-qubit level. The CPW linewidth is therefore far above the modeled single-qubit retention target; this comparison is a literature-based gap analysis, not a universal platform ranking. The experimental program in the main text is staged to first establish the loss-controlled benchmark on a 3D cavity, then extend to collective coupling on a CPW-style architecture, and finally combine the two. None of the three stages has been demonstrated end-to-end.

The protocol suggests platform candidates, but implementation requires a device-specific validation of coupling uniformity, modulation fidelity, decoherence, and readout. **Flux**

qubits are the original platform for USC [12], with tunable coupling via SQUID loops, best for $N = 1$ to 4. **Transmons** are less anharmonic but more coherent, and can reach USC with flux-tunable couplers [11], best for $N > 8$ due to fabrication uniformity. **Fluxonium** offers high anharmonicity and long coherence, promising for USC with $N = 1$ to 2 [14]. Platform-specific circuit schematics are provided in Section S8.

We recommend a staged experimental approach:

1. **Stage** 1 ($N = 1$). Benchmark the Rabi+DCE protocol against a matched pump-power and loss reference. The calculated $170\,903\times$ peak is a pure-state model value; reaching it experimentally requires validation of the USC operating point, drive calibration, and readout. [12, 13].

2. **Stage** 2 ($N = 4$ to 8). Test collective scaling in a small array while measuring coupling inhomogeneity, thermal noise, and the actual pump cost. The quoted $N = 8$ factors remain model predictions.

3. **Stage** 3 ($N = 16$ to 32). Test the full Dicke+DCE parameter set only after the preceding calibration and matched-resource gates pass. Uniform coupling to approximately 32 qubits and the required high-fidelity modulation remain open feasibility questions.

S11. LOSS-ADJUSTED DCE QFI UNDER CAVITY DISSIPATION

To assess the robustness of the DCE QFI against cavity loss, we perform Lindblad master-equation simulations of the mixed-state QFI at the published operating point ($\eta = 0.150$, $g_d/g_0 = 0.119\,23$, $\omega_q/\omega_c = 1.027$, $\omega_d/\omega_c = 2.077$). The Hamiltonian matches the original scan script exactly: $H(t) = \frac{1}{2}\omega_q\sigma_z + \omega_c a^\dagger a + (\eta + g_d\cos\omega_d t)\,\sigma_x(a + a^\dagger)$, with bare vacuum initial state $|0\rangle \otimes |\downarrow\rangle$, Fock truncation $N_{\text{cav}} = 20$, and fidelity-based QFI $F_Q = 8(1 - \sqrt{F})/\delta^2$ with $\delta = 1 \times 10^{-3}$. The collapse operator is $\sqrt{\kappa}\,a$ (single-photon cavity loss); no qubit dissipation is included.

Table S9 reports the peak and late-window QFI for each κ/ω_c. At the simulated physical operating point $\kappa/\omega_c \approx 4.1 \times 10^{-2}$ ($\kappa/2\pi \approx 246\,\text{MHz}$; the cited CPW reference is approximately 248 MHz), the peak QFI retains only $\sim 1\%$ of the pure-state value, far below the simple exponential envelope $\exp(-\kappa t)$ because the Lindblad dynamics modify the entire

state evolution rather than merely attenuating the peak amplitude. The degradation becomes severe for $\kappa/\omega_c \gtrsim 1 \times 10^{-3}$. In the sampled sweep, the first row at or above 80 % retention is $\kappa/2\pi = 0.6$ MHz (80.6%); because the lower-loss rows are not strictly monotonic, no continuous cutoff is inferred.

TABLE S9. Loss-adjusted DCE QFI under single-photon cavity loss. Peak QFI and late-window average (last 5 drive periods) at the published operating point. The reference pure-state peak is 170 903.

κ/ω_c	$\kappa/2\pi$ (MHz)	Peak QFI	Late-window avg QFI	% of pure (peak)
0	0	170 903	132 800	100
1×10^{-5}	0.06	133 806	100 003	78
5×10^{-5}	0.3	134 693	100 977	79
1×10^{-4}	0.6	137 772	104 482	81
5×10^{-4}	3	119 926	96 606	70
1×10^{-3}	6	82 687	71 075	48
5×10^{-3}	30	11 653	6287	6.8
1×10^{-2}	60	5957	1919	3.5
2×10^{-2}	120	3376	853	2.0
4.1×10^{-2}	246	1640	406	0.96

S12. TIME-STEP AND FINITE-DIFFERENCE CONVERGENCE OF DCE QFI

We verify the numerical stability of the published DCE QFI (170 903×) against both the integration time step and the finite-difference parameter δ. The test uses the same Hamiltonian, initial state, and parameters as Section S11.

S12.1. Time-step convergence

Table S10 reports the peak QFI as a function of the number of time steps n_{steps} over 40 drive periods, with $\delta = 1 \times 10^{-3}$ fixed. The peak QFI varies by $< 0.15\%$ across $n_{\text{steps}} = 100$–2000, confirming that the published value ($n_{\text{steps}} = 500$, $\Delta t/T_d = 0.08$) is converged.

TABLE S10. DCE QFI convergence with time-step refinement.

n_{steps}	$\Delta t/T_d$	Peak QFI	Relative to 500
100	0.400	170 671	−0.14%
200	0.200	170 667	−0.14%
500	0.080	170 903	—
1000	0.040	170 687	−0.13%
2000	0.020	170 687	−0.13%

S12.2. Finite-difference step convergence

Table S11 reports the peak QFI as a function of δ with $n_{\text{steps}} = 500$ fixed. The published value $\delta = 1 \times 10^{-3}$ gives 170 903×; refining to $\delta = 1 \times 10^{-4}$ changes the peak by $< 0.12\%$, confirming that the finite-difference approximation is well within the converged regime.

TABLE S11. DCE QFI convergence with finite-difference step δ.

δ	Peak QFI	Relative to $\delta = 1 \times 10^{-3}$
1×10^{-2}	72 943	−57%
5×10^{-3}	153 912	−10%
1×10^{-3}	170 903	—
5×10^{-4}	171 579	+0.40%
1×10^{-4}	171 093	+0.11%

S13. RESOURCE-NORMALIZED DCE QFI BENCHMARKS

To address whether the DCE QFI advantage is real or an artifact of an unconstrained pump, we sweep the modulation amplitude g_d at fixed $\eta = 0.150$ and compute (i) the QFI per pump-photon proxy g_d^2 (time-averaged modulation energy), (ii) the QFI per intracavity photon n_{peak} and n_{avg}, and (iii) the squeezed-state QFI at the same intracavity photon budget. The pure-state QFI at the published optimum is 170 903× at $n_{\text{peak}} = 9.01$ and $n_{\text{avg}} = 5.72$.

Table S12 reports the scan. The ratio to the pump-energy proxy is largest at the smallest g_d tested ($g_d = 0.05$, QFI/pump $\sim 3.7 \times 10^7$), but the absolute QFI is smaller there; this proxy is not a calibrated pump-power cost. The QFI per intracavity photon is largest at the published $g_d = 0.119\,23$ within this scan, giving $29\,891\times$ per average photon; larger g_d increases the modeled photon population without a commensurate QFI increase in the sampled points. The squeezed-state reference at matched photon budget $n = 9.01$ gives $F_Q^{\mathrm{sq}} = 4n(1{+}2n) = 685.7\times$, numerically below the DCE peak in this formal comparison. This establishes only a formal comparison at equal intracavity population. It does not establish an operational advantage because the pump, bandwidth, preparation, loss, readout, and observation-time costs are not matched.

TABLE S12. Resource-normalized DCE QFI benchmarks at $\eta = 0.150$.

g_d	g_d^2	QFI peak	n_{peak}	n_{avg}	QFI / n_{avg}
0.05	0.0025	91 738	3.56	1.64	55 831
0.10	0.0100	146 863	8.49	4.73	31 031
0.119 23	0.014 22	170 903	9.01	5.72	29 891
0.15	0.0225	166 990	6.39	6.23	26 822
0.20	0.0400	105 529	4.29	6.04	17 474
0.30	0.0900	40 154	7.34	8.17	4917
0.50	0.2500	6131	2.70	2.96	2074

At matched photon budget $n = 9.01$, the coherent-state shot-noise reference is $F_Q^{\mathrm{SQL}} = 4n = 36.0$, the stated pure squeezed-state reference is $F_Q^{\mathrm{sq}} = 4n(1 + 2n) = 685.7$, and the DCE peak is $F_Q^{\mathrm{DCE}} = 170\,903$. At $n = 5.72$, the DCE time-average photon budget, the corresponding reference values are listed in the data artifact. These quantities are all idealized pure-state or reference-model values. Photon-number matching alone does not match pump power, bandwidth, preparation, loss, readout, or observation time, so the ratios are not device-level advantages. Under cavity loss the DCE QFI collapses (Section S11); the competing probe must be propagated under the same estimation task and loss model before any operational comparison can be made.

S14. ADAPTIVE PROTOCOL COMPARISON AT MATCHED PHOTON BUDGET

The static coherent-state shot-noise reference is $F_Q^{\mathrm{SQL}} = 4n$. Adaptive phase-estimation analyses provide an uncertainty-bound reference approaching $\sim 1/n$ for known amplitude and optimal-feedback assumptions (Wiseman-Doherty). This is an uncertainty bound, not a QFI value, and it is not directly comparable to the dimensionless DCE QFI. The single-mode expression $(4n + 1)/n^2$ is retained as a formal Helstrom reference under its stated known-reference assumptions, not as an adaptive QFI.

Table S13 reports the formal references at the DCE peak photon budget and at the DCE time-average photon budget. The $n = 5.72$ row reuses the DCE peak QFI as a labeled reference at the time-average photon budget; it is not a matched-probe calculation. Because the adaptive entries are uncertainty-bound or formal reference expressions whereas the DCE entries are QFI values, no cross-estimand enhancement ratios are reported. A physically relevant comparison would require a common estimation task, prior, observation time, loss model, drive budget, and readout; that comparison is outside the present study. The DCE protocol's explicit cavity-loss sensitivity is quantified in Section S11.

TABLE S13. Adaptive protocol comparison at matched intracavity photon budget.

n	SQL QFI $(4n)$	Adaptive uncertainty bound $(1/n)$	Formal reference	DCE QFI reference
5.72	22.9	0.175	0.73	170 903
9.01	36.0	0.111	0.46	170 903

S15. LOSS-ADJUSTED DICKE+DCE QFI (N=32)

To extend the loss analysis to the collective Dicke+DCE protocol, we performed Lindblad master-equation simulations of the mixed-state QFI at the published Dicke+DCE operating point ($N = 32$, $\eta = 1.15$, $g_d/g_0 = 0.15$, $\omega_q/\omega_c = 1.0$, $\omega_d/\omega_c = 2.0$). The Hilbert space is $N_{\mathrm{cav}} \times (2J + 1) = 6 \times 33 = 198$, with photon truncation reduced from the published $N_{\mathrm{cav}} = 28$ to enable the Lindblad propagation at tractable cost. The absolute QFI and loss curve are therefore truncation-dependent and are not treated as converged. The initial state

is the bare Dicke ground state $|0\rangle_{\mathrm{cav}} \otimes |J, -J\rangle$ (Hilbert-space dim 198). The Hamiltonian is $H(t) = \omega_c a^\dagger a + \omega_q J_z + (g_0 + g_d \cos \omega_d t)/\sqrt{N}\, J_x(a + a^\dagger)$ with $g_0 = \eta\, \omega_c$.

Table S14 reports the peak QFI for each κ/ω_c at the validated $N_{\mathrm{cav}} = 6$ truncation. All values come from the validated 5-point κ-sweep (array jobs 15744/15764/15784), whose per-point output is preserved in the local SLURM logs `slurm-15744_*.out`, `slurm-15764_*.out`, and `slurm-15784_*.out`; the checkpoint JSONs were pruned after the canonical merge. The adaptive integrator runs at `atol` = `rtol` = 1×10^{-10} with $n_{\mathrm{steps}} = 200$ over ten drive periods, and every stored state passes the trace, Hermiticity, and numerical eigenvalue acceptance checks (worst min eigenvalue -1.3×10^{-5} against the $-1e-4$ acceptance floor); this is not a proof of exact positivity. The top-level validation status is recorded in the canonical JSON `validation` block and the DAR. The pure-state peak (at the reduced $N_{\mathrm{cav}} = 6$ and $T_{\max} = 10$ periods) is 2962× — considerably below the published 251 557× at $N_{\mathrm{cav}} = 28$ and $T_{\max} = 40$ periods, reflecting the truncation. The relative retention under cavity loss is the operationally meaningful number: the $N_{\mathrm{cav}} = 6$ Dicke+DCE QFI retains 99 % at $\kappa/\omega_c = 1 \times 10^{-4}$, 95 % at 1×10^{-3}, 61 % at 1×10^{-2}, and 16 % at the physical point $\kappa/\omega_c = 4.1 \times 10^{-2}$ ($\kappa/2\pi \simeq 246$ MHz). The physical-point entry is now reported because the propagation passes the numerical physicality gate; an earlier estimate of $\sim$ 15 % retention at $\kappa/\omega_c = 1 \times 10^{-2}$ and a claimed numerical-noise floor at the physical point came from default-tolerance integration with min eigenvalues down to -4.8×10^{-4} and are superseded by these validated values.

Truncation is *not* converged within the validated reduced-truncation window. Intermediate recomputations at $N_{\mathrm{cav}} = 10$ and 12 (artifacts `data/dicke_dce_loss_adjusted_n10.json` and `data/dicke_dce_loss_adjusted_n12.json`, same solver settings as the $N_{\mathrm{cav}} = 6$ sweep) show that the pure-state reference QFI and the loss retention are both strongly truncation-dependent: the $N_{\mathrm{cav}} = 6$ pure-state peak (2962×) is above the $N_{\mathrm{cav}} = 10$ value (2363×) but below $N_{\mathrm{cav}} = 12$ (4833×), and the retention at the physical κ decreases from 16 % at $N_{\mathrm{cav}} = 6$ to 12 % at $N_{\mathrm{cav}} = 10$ and 7 % at $N_{\mathrm{cav}} = 12$ (Table S15). The truncation extension at $N_{\mathrm{cav}} \in \{16, 20, 24\}$ (artifact `data/dicke_dce_loss_truncation_extension.json`, status `partial_or_inconclusive`) extends this trend: the same-truncation zero-loss reference at $N_{\mathrm{cav}} = 16$ gives a peak QFI of 5785.6×, at $N_{\mathrm{cav}} = 20$ of 9851.1×, and at $N_{\mathrm{cav}} = 24$ of 10 583.6×; under cavity loss the QFI retention falls to 32.7% at $\kappa/\omega_c = 1 \times 10^{-2}$ and to 4.4% at the physical point for $N_{\mathrm{cav}} = 16$, to 24.1% and 2.7% at the same points for $N_{\mathrm{cav}} = 20$, and

to 17.0% and 2.6% at the same points for $N_{\mathrm{cav}} = 24$. The $N_{\mathrm{cav}} = 16 \to 24$ relative peak-QFI differences are 45.3 % at $\kappa = 0.0$, 4.9 % at $\kappa/\omega_c = 1 \times 10^{-2}$ (within the declared 5% tolerance), and 5.9 % at $\kappa/\omega_c = 4.1 \times 10^{-2}$; the $N_{\mathrm{cav}} = 20 \to 24$ differences are 6.9 % at $\kappa = 0.0$, 31.6 % at $\kappa/\omega_c = 1 \times 10^{-2}$, and 2.9 % at $\kappa/\omega_c = 4.1 \times 10^{-2}$ (within tolerance). Two of the six truncation comparisons ($16 \to 24$ at $\kappa/\omega_c = 1 \times 10^{-2}$ and $20 \to 24$ at $\kappa/\omega_c = 4.1 \times 10^{-2}$) meet the tolerance criterion, so the extension status remains `partial_or_inconclusive`. The physical-point retention is therefore a truncation-dependent range of roughly 2.6%–16 % over $N_{\mathrm{cav}} \in \{6, 10, 12, 16, 20, 24\}$, not a truncation-converged number, and no Dicke-specific platform threshold is claimed. The reduced-truncation diagnostics at $N_{\mathrm{cav}} = 6$ place the QFI peak at approximately $9.65\,T_d$ for $\kappa/\omega_c \leq 1 \times 10^{-3}$, at $9.60\,T_d$ for $\kappa/\omega_c = 1 \times 10^{-2}$, and at $6.13\,T_d$ at the physical point $\kappa/\omega_c = 4.1 \times 10^{-2}$. This loss-induced shift is a diagnostic of the reduced ten-period propagation, not evidence of truncation convergence.

TABLE S14. Loss-adjusted Dicke+DCE QFI under single-photon cavity loss ($N = 32$, $\eta = 1.15$, $g_d/g_0 = 0.15$, $N_{\mathrm{cav}} = 6$ for tractability, $T_{\max} = 10$ drive periods). Validated checkpoint values (2026-09-02).

κ/ω_c	$\kappa/2\pi$ (MHz)	Peak QFI	Late-window avg QFI	Retained
0	0	2962	2584	
1×10^{-4}	0.6	2927	2566	99 %
1×10^{-3}	6	2828	2480	95 %
1×10^{-2}	60	1818	1678	61 %
4.1×10^{-2}	246	474	321	16 %

Validated Lindblad propagation (atol = rtol = 1×10^{-10}, nonnormal eigensolver acceptance floor $-1e-4$; not a proof of exact positivity). Retention is relative to the $N_{\mathrm{cav}} = 6$ pure-state peak and is truncation-dependent. The $N_{\mathrm{cav}} = 10$ and 12 intermediate recomputations (`data/dicke_qfi_vs_kappa.json`, `data/dicke_qfi_vs_kappa_updated.json`) and the truncation extension at $N_{\mathrm{cav}} \in \{16, 20, 24\}$ (artifact `data/dicke_qfi_vs_kappa_extension.json`) are reported in Table S15. The reduced-truncation late-window average is not interpreted as a steady-state quantity.

Table S15 summarizes the truncation dependence across $N_{\mathrm{cav}} = 6$–24 at the physical $\kappa/\omega_c = 4.1 \times 10^{-2}$. The pure-state reference QFI is non-monotonic with N_{cav} (the $N_{\mathrm{cav}} = 6$ value is above $N_{\mathrm{cav}} = 10$ due to Fock-space confinement), but the retention under cavity loss decreases monotonically, confirming that larger truncations capture more loss channels.

TABLE S15. Truncation dependence of Dicke+DCE QFI retention at the physical cavity linewidth. Retention is relative to each truncation's zero-loss peak; all rows use the same solver settings. See the text and the stored JSON artifact for provenance and the pairwise 5% convergence results.

N_{cav}	Hilbert dim	Peak QFI ($\kappa = 0$)	Peak QFI (physical κ)	Retention
6	198	2962	474	16 %
10	330	2363	290	12 %
12	396	4833	320	7 %
16	528	5786	255	4.4 %
20	660	9851	265	2.7 %
24	792	10 584	273	2.6 %

Same solver settings: `atol` = `rtol` = 1×10^{-10}, $n_{\mathrm{steps}} = 200$, $T_{\max} = 10$ drive periods, eigenvalue floor $-1e-4$. Retention is truncation-dependent and not converged.

S16. LOSS-ADJUSTED DCE QFI WITH COMBINED CAVITY + QUBIT DISSIPATION

The cavity-only Lindblad analysis in Section S11 is incomplete: physical USC platforms also exhibit qubit relaxation (T_1) and pure dephasing (T_ϕ) at rates that are non-negligible over the simulated window. We extend the Lindblad master equation to include both effects.

For the conservative transmon-like benchmark at $\omega_c/2\pi = 6\,\mathrm{GHz}$ ($T_1 = 30\,\mu\mathrm{s}$, $T_\phi = 5\,\mu\mathrm{s}$), the SI-derived dimensionless rates are $\gamma_1/\omega_c = 8.84 \times 10^{-7}$ and $\gamma_\phi/\omega_c = 5.31 \times 10^{-6}$. These rates are computed from the declared coherence times and the angular cavity frequency; they are not state-of-the-art-device parameters. The cavity-loss rate at the physical CPW operating point is $\kappa/\omega_c = 4.1 \times 10^{-2}$.

Table S16 reports the peak QFI under each dissipation channel individually and combined. The cavity loss at the physical CPW rate ($\kappa/\omega_c = 4.1 \times 10^{-2}$) reduces the QFI to approximately 0.96% of the pure-state value. With no cavity loss, qubit T_1 alone retains 78.56%, pure dephasing alone retains 79.08%, and both qubit channels together retain 79.08%. Adding both qubit channels at the physical cavity-loss point gives 0.9594%, essentially the cavity-only value; cavity decay is therefore the dominant channel over the 40-period window. At the sampled $\kappa/2\pi = 0.6\,\mathrm{MHz}$ point, adding the T_1 benchmark retains approximately 80.6% of the pure-state QFI. These are conservative, model-specific benchmark results, not device-performance predictions.

The qubit-loss contribution is non-negligible but not dominant in this modeled window. The cavity-loss barrier remains the principal constraint in the tested parameter set; reducing κ to the 3D-cavity range ($\kappa/2\pi \lesssim 1\,\mathrm{MHz}$) increases the modeled DCE retention, but does not by itself establish an operational advantage or guarantee performance for another qubit platform. For flux qubits, which are the historical USC platform but have historically reported shorter relaxation times, $T_1 \sim 1\,\mu\mathrm{s}$ to $10\,\mu\mathrm{s}$ in early USC-era devices (factor 3 to 30 shorter than the transmon value adopted here), the qubit-loss contribution could be larger for those devices; redesigned flux qubits have since reported T_1 in excess of $40\,\mu\mathrm{s}$ at the flux-insensitive point [16], so this shorter range is not a fixed platform ceiling. Neither range was fully parameterized in the present study.

TABLE S16. Loss-adjusted DCE QFI under combined cavity and qubit dissipation. Pure-state reference: 170 903.

Configuration	κ/ω_c	γ_1/ω_c	γ_ϕ/ω_c	Peak QFI
Pure (no loss)	0	0	0	170 903
Cavity only ($\kappa = 1 \times 10^{-4}$)	1×10^{-4}	0	0	137 772
Cavity only ($\kappa = 1 \times 10^{-2}$)	1×10^{-2}	0	0	5957
Cavity only ($\kappa = 4.1 \times 10^{-2}$, physical)	4.1×10^{-2}	0	0	1640
Qubit $T_1 = 30\,\mu$s only	0	8.84×10^{-7}	0	134 258
Qubit $T_\phi = 5\,\mu$s only	0	0	5.31×10^{-6}	135 156
$T_1 + T_\phi$ (no cavity)	0	8.84×10^{-7}	5.31×10^{-6}	135 147
Cavity + qubit (physical CPW)	4.1×10^{-2}	8.84×10^{-7}	5.31×10^{-6}	1640
3D cavity ($\kappa = 1 \times 10^{-4}$) + qubit T_1	1×10^{-4}	8.84×10^{-7}	0	137 769

S17. DICKE+DCE BOUNDARY PEAK: CONVERGENCE ANALYSIS

The Dicke+DCE QFI in the main text was reported with two values: a late-window average (167 271×, 730× enhancement) and a boundary peak at the last stored time point (251 557×, 1098× enhancement). The boundary peak caveat is important: the published data ends at drive period 40 and may not represent a converged local maximum. To test this, we extended the pure-state Dicke+DCE time trace (at the same $N_{\mathrm{cav}} = 6$ truncation used in the loss analysis) to drive periods 20 to 60 and report the peak QFI at each.

Table S17 reports the result. Within the tested windows, the peak QFI grows monotonically with $T_{\max}$: at period 20 the peak is 9932×, at period 40 it is 33 109× (3.3× higher), and at period 60 it is 69 475× (7.0× higher than period 20). The peak occurs at the *last* stored time point for every $T_{\max}$ tested, indicating that the QFI is still rising at the boundary of the simulation window. The published 251 557× at $N_{\mathrm{cav}} = 28$ and $T_{\max} = 40$ is therefore an arbitrary boundary value, not a physical saturation; the calculation does not establish the long-time behavior; extending the simulation window is required before any saturation claim can be made.

This reinforces the operational message: within the tested pure-state windows, the Dicke+DCE QFI remains a finite-window upper-bound estimate rather than a converged

TABLE S17. Dicke+DCE QFI boundary-peak convergence ($N = 32$, $\eta = 1.15$, $g_d/g_0 = 0.15$, $N_{\mathrm{cav}} = 6$ for tractability). The peak occurs at the last stored time point for every $T_{\max}$, indicating a still-rising boundary.

$T_{\max}$ (periods)	Peak QFI	Late-window avg	Peak at boundary?
20	9932	8219	yes
30	19 093	16 060	yes
40	33 109	27 535	yes
50	48 328	39 760	yes
60	69 475	57 117	yes

saturation. The late-window average (730× enhancement) is the operationally meaningful figure, and cavity loss removes most of it even within the validated $T_{\max} = 10$-period reduced-model window (retention falls to $\sim 61\,\%$ at $\kappa/\omega_c = 1 \times 10^{-2}$ and $\sim 16\,\%$ at the physical κ at $N_{\mathrm{cav}} = 6$, decreasing through $\sim 12\,\%$ at $N_{\mathrm{cav}} = 10$ and $\sim 7\,\%$ at $N_{\mathrm{cav}} = 12$ to $\sim 3\,\%$–$5\,\%$ at $N_{\mathrm{cav}} \in \{16, 20\}$ and $\sim 2.6\,\%$ at $N_{\mathrm{cav}} = 24$ in the partial-converged extension (two of six truncation comparisons meet the 5% tolerance: $16 \to 24$ at $\kappa/\omega_c = 1 \times 10^{-2}$ and $20 \to 24$ at the physical κ; Section S15). The published boundary peak $2.5 \times 10^5\times$ should therefore be treated only as a finite-window diagnostic, not as a quantitative saturation result.

S18. MODEL-SPECIFIC INTERROGATION-WINDOW ANALYSIS

To test whether cavity loss changes the useful interrogation window, we propagated the single-qubit DCE protocol with the mixed-state Lindblad master equation at the fixed operating point $\eta = 0.150$, $g_d/g_0 = 0.119\,23$, $\omega_q/\omega_c = 1.027$, and $\omega_d/\omega_c = 2.077$. The cavity truncation is $N_{\mathrm{cav}} = 20$ and the window is 40 drive periods. For each κ/ω_c, we calculate the fidelity-based mixed-state QFI $F_Q(t)$, the von Neumann entropy $S(\rho_t)$ in bits, and the declared information-rate objective $R_Q(t) = F_Q(t)/t$ after the first drive period. The reference and perturbed trajectories use the same loss channel and differ by $\delta\eta = 1 \times 10^{-3}$.

The result is a model-specific crossover rather than a thermodynamic law. At $\kappa/\omega_c \leq 1\times10^{-3}$, both the QFI and R_Q reach their largest stored values at the 40-period boundary. At

larger loss, the maximum of R_Q moves into the window: it occurs at 18.08, 14.00, 10.96, and 6.96 periods for $\kappa/\omega_c = 0.005$, 0.01, 0.02, and 0.041, respectively. At these information-rate maxima, the QFI is 10 448.8, 5510.1, 2960.5, and 1242.2, while the entropy is 1.572, 1.582, 1.599, and 1.350 bits. These values indicate that dissipation can shift the best information-acquisition rate earlier, but they do not define a universal optimal interrogation time.

The entropy derivative stored in the artifact is a finite-difference diagnostic of state mixedness. It is not the entropy-production rate of a specified thermodynamic environment: no heat/work decomposition, bath entropy, Spohn production, output-field record, detector efficiency, or pump-power cost is included. The calculation also uses a finite time window and a single truncation. Accordingly, item 3.6 is promoted only to a qualified, model-specific information-rate analysis; the thermodynamic-optimum and resource-complete versions remain open.

TABLE S18. Model-specific information-rate maxima for the single-qubit DCE protocol. The objective is $R_Q = F_Q/t$ after one drive period; t is normalized inverse-cavity-frequency time and the reported period is t/T_d.

κ/ω_c	$\kappa/2\pi$ (MHz)	t/T_d at R_Q max	F_Q at R_Q max	S (bits)	Window status
0	0	40.00	170 670	0.000	boundary
1×10^{-4}	0.6	40.00	161 087	0.454	boundary
1×10^{-3}	6	40.00	84 830	2.005	boundary
5×10^{-3}	30	18.08	10 449	1.572	interior
1×10^{-2}	60	14.00	5510	1.582	interior
2×10^{-2}	120	10.96	2961	1.599	interior
4.1×10^{-2}	246	6.96	1242	1.350	interior

S19. FINITE-WINDOW FLOQUET STABILITY SCREEN

The proposed passive-stabilizer interpretation is tested here as a finite-window numerical screen rather than assumed from the static energy budget. The producer `scripts/floquet_stability.py` evaluates the driven single-qubit Rabi model with $\omega_q/\omega_c = 1.027$, coupling modulation $g(t) = g_0[1 + (g_d/g_0)\cos(\omega_d t + \phi)]$, and cavity loss

κ. The coupling-estimand QFI is obtained from paired trajectories at η and $\eta + \delta\eta$, with $\delta\eta = 1 \times 10^{-3}$. Tier 1 uses $N_{\mathrm{cav}} = 12, 20$ and 40 drive periods over 108 parameter rows; the selected Tier 2 validation uses 80 periods and $N_{\mathrm{cav}} = 20, 30$ for five representative rows. Zero-loss rows use unitary `sesolve`; lossy rows use Lindblad `mesolve` with the single-photon cavity collapse operator $\sqrt{\kappa}\, a$.

All 432 Tier 1 trajectories pass the numerical trace, Hermiticity, and eigenvalue acceptance gate, as do the Tier 2 trajectories. The Tier 1 screen classifies 79 rows as showing 'growth_at_boundary' and 29 as 'bounded_in_screen'; no row exceeds the declared 10% truncation-difference criterion. The Tier 2 resonant/low-loss rows continue to grow at the 80-period boundary, whereas the two selected lossy rows remain bounded in that finite window and agree between $N_{\mathrm{cav}} = 20$ and 30 to better than 1×10^{-6} in peak QFI.

These results do not establish asymptotic Floquet stability, suppression of parametric divergence, or phase stability. In particular, the persistence of boundary growth in selected low-loss resonant rows rejects the claim that two-level saturation is a general passive Floquet stabilizer. The bounded rows support only a model-specific statement about the tested finite window, loss, drive, initial state, truncation, and solver settings. The complete row-level diagnostics are stored in `data/floquet_stability.json` and `data/floquet_stability_validation.json`.

TABLE S19. Summary of the finite-window Floquet stability screen. 'Growth' denotes a photon-number or QFI boundary-growth flag; 'bounded' means no such flag in the stated window.

Screen	Rows	N_{cav}	Periods	Result
Tier 1	108	12,20	40	79 growth; 29 bounded
Tier 2 selected	5	20,30	80	3 growth; 2 bounded

S20. MULTIPARAMETER QFI AND REPRESENTATION CONSISTENCY

To separate sensitivity to the coupling from sensitivity to nuisance parameters, we evaluated the normalized parameter vector $(\eta, \omega_q/\omega_c, g_d/g_0, \phi)$ for the driven protocol and $(\eta, \omega_q/\omega_c)$ for static ground states. The producer `scripts/multiparameter_gauge_qfi.py` uses the existing tensor ordering and QuTiP `sesolve` convention. It evaluates pure-state

QFI metric tensors from phase-aligned central finite differences at steps 1×10^{-2}, 1×10^{-3}, and 1×10^{-4}, and stores eigenvalues, normalized correlations, rank, condition number, and the effective information for η after projecting out the other declared parameters.

The artifact `data/multiparameter_gauge_qfi.json` contains 18 rows: static points at $\eta = 0.10$, 0.15, and 1.00 for $N_{\mathrm{cav}} = 12, 20$, and driven points at relative frequency offsets -0.02, 0, and $+0.02$ over 6 and 10 drive periods at the same truncations. All rows are finite, positive semidefinite within the numerical tolerance, and classified as converged under the stored step-refinement criterion. At the canonical 10-period row and $N_{\mathrm{cav}} = 20$, the coupling diagonal is approximately 1.15×10^3 and the nuisance-projected effective coupling information is approximately 4.08×10^2. The driven condition numbers are of order 1×10^3, so directional estimates remain numerically ill-conditioned even when the step-refinement gate passes.

As a controlled representation test, we apply $U = \exp[-i0.37a^\dagger a]$ at $N_{\mathrm{cav}} = 30$ and transform the Hamiltonian, initial state, control interaction, and observables together. The unitarity error is 1.6×10^{-15}, the transformed-state infidelity is below 4×10^{-16}, the largest recorded internal-observable discrepancy is below 2.2×10^{-12}, and the relative QFI-metric difference is 4.8×10^{-9}. This verifies the implemented finite-dimensional similarity transformation. It is not a comparison of arbitrary laboratory gauges with independently specified controls, and it does not establish phase-noise protection or device-level gauge independence.

TABLE S20. Selected model-specific multiparameter QFI diagnostics from the stored artifact. The QFI entries use normalized parameters; the phase parameter is in radians.

Row	N_{cav}	$F_{\eta\eta}$	$F_{\eta\eta}^{\mathrm{nuisance}}$	Condition number
Static, $\eta = 0.10$	20	1.003	0.004 92	8.63×10^4
Static, $\eta = 1.00$	20	5.766	1.151	175
Driven, $\Delta\omega_d = 0$, $6T_d$	20	328.604	120.098	1.10×10^3
Driven, $\Delta\omega_d = 0$, $10T_d$	20	1148.943	408.475	793

S21. OUTPUT-FIELD HOMODYNE CFI SCREEN

The idealized joint-system QFI does not by itself specify a laboratory readout. We therefore evaluated a bounded output-field measurement model using the input–output convention

$$a_{\mathrm{out}}(t) = a_{\mathrm{in}}(t) + \sqrt{\kappa_{\mathrm{ext}}}\, a(t), \tag{S16}$$

with vacuum input and $\kappa = \kappa_{\mathrm{ext}} + \kappa_{\mathrm{int}}$. Only κ_{ext} is treated as monitored output; the internal-loss channel is unobserved. The measured temporal mode is flat, $f(t) = 1/\sqrt{\tau}$, over a window of $\tau = 6$ or 10 drive periods. The quadrature is $X_\theta = (A_f e^{-i\theta} + A_f^\dagger e^{i\theta})/\sqrt{2}$ with vacuum shot-noise variance 1/2.

The producer `scripts/homodyne_cfi.py` uses QuTiP `mesolve` and quantum-regression two-time correlations, followed by composite-trapezoid temporal-mode integration. It calculates the CFI of a Gaussian quadrature approximation from the output-mode mean and variance, not the exact generally non-Gaussian temporal-mode distribution. The artifact `data/homodyne_cfi.json` contains 24 resumable parameter groups and 2304 rows spanning static and driven protocols, $\kappa/\omega_c = 0$, 0.01, and 0.041, external fractions 0.5 and 1, detector efficiencies 0.5, 0.8, and 1, four LO phases, mode resolutions 8 and 12, $N_{\mathrm{cav}} = 12, 20$, and coupling steps $\delta\eta = 1 \times 10^{-3}$ and 5×10^{-4}. The output-mode Gaussian QFI comparator is formed from the same nominal, positive, and negative output-mode moments; the residual joint cavity–qubit QFI is recorded separately.

All quadrature variances and CFI values are finite and non-negative. Among rows with a valid same-state comparison, $0 \le F_C/F_Q \le 0.603$; rows with zero or numerically invalid output-mode QFI are retained as inconclusive rather than clipped. The largest valid ratio is 0.603 (60.3 %) for a driven row in the tested grid, while the largest static ratio is 0.0455 (4.55 %). The historical 92.4 % static and 88.1 % driven hypotheses are both classified `not_reproduced` in the JSON artifact.

This result supports only a model-specific Gaussian homodyne-CFI screen under the stated loss split, flat temporal mode, detector efficiency, finite window, and cavity truncation. It does not establish exact field CFI, matched-filter optimality, experimental QFI saturation, universal homodyne optimality, or resource-normalized performance.

TABLE S21. Selected output-field homodyne-CFI results. Ratios use the matching reconstructed Gaussian output-mode QFI and are not experimental saturation fractions.

Protocol	κ/ω_c	ϵ_d	F_C/F_Q	Status
Static	0.01	1.0	0.000 484	valid
Static	0.041	1.0	0.002 63	valid
Driven	0.01	1.0	0.0129	valid
Driven	0.041	1.0	0.603	valid

S22. HEOM–LINDBLAD CROSS-VALIDATION

We use a local Spin–Boson comparison to assess the bath-model approximation behind the Lindblad loss calculations. The stored artifact `data/trackB_heom_validation.json` uses a Drude–Lorentz bath with normalized parameters $\omega_q = 1.0$, $\gamma_{\mathrm{cut}} = 20$, $\lambda = 0.02$, $\Gamma = 0.012\,535$, $T_{\mathrm{bath}} = 0.001$, hierarchy depth 3, and 50 time points to $t_{\mathrm{final}} = 1.0$ in the stored units. It reports a mean Uhlmann-fidelity error of 1.18%, a maximum of 2.43%, and all sampled points below 5%.

This is a short-time, local bath-model cross-check, not a convergence proof for the hierarchy, not a validation of the driven high-QFI traces, and not a validation of the full mixed-state metrological protocol. In particular, the stored metadata do not encode the proposed $4\pi/\omega_c$ interrogation interval or a sweep over the DCE operating parameters. The comparison therefore supports using Lindblad dynamics as the working model for the reported loss studies within their stated scope, while leaving memory corrections outside that scope unresolved.

TABLE S22. Stored local HEOM–Lindblad cross-check.

Metric	Stored value
Mean fidelity error	1.18%
Maximum fidelity error	2.43%
All sampled points below 5%	yes
Hierarchy depth	3
Stored final time	1.0

S23. FINITE-TEMPERATURE RETENTION SCREEN

To test how finite temperature alters the virtual-state and driven QFI, the producer `scripts/thermal_competition.py` propagates the canonical static-Rabi and driven single-qubit DCE models under a finite-temperature Lindblad master equation. The cavity is coupled to explicit emission and absorption channels $L_- = \sqrt{\kappa(n_{\rm th}+1)}\,a$ and $L_+ = \sqrt{\kappa\, n_{\rm th}}\,a^\dagger$, with the Bose–Einstein occupation $n_{\rm th} = [\exp(\hbar\omega_c/k_BT) - 1]^{-1}$ evaluated at the declared physical cavity frequency. Separate qubit-thermalization rows add the qubit relaxation channel at its own transition frequency and temperature. The grid spans $T \in [10, 200]$ mK, $\omega_c/2\pi \in [5, 8]$ GHz, static rows at $\eta = 0.15$ and 1.0, driven rows at the published operating point ($\eta = 0.150$, drive coefficient $g_d = 0.119\,23$ in the shared normalized convention, $\omega_q/\omega_c = 1.027$, $\omega_d/\omega_c = 2.077$), $\kappa/\omega_c = 1 \times 10^{-3}$ and 4.1×10^{-2}, and three declared initial states (bare thermal, dressed, vacuum). All 115 stored rows pass the numerical trace, Hermiticity, and eigenvalue acceptance gate.

The retention criterion is declared before scanning: the first sampled temperature at which the peak QFI relative to the same-configuration zero-temperature reference falls below 80 %. Of the 17 recorded reference families, 12 reach a sampled crossover within 200 mK; five families retain at least 80 % through 200 mK. Table S23 lists the canonical driven row: its zero-temperature peak QFI is 1642.1 (matching the interrogation artifact to the stored precision), retention is 0.985 at 50 mK and 0.775 at 100 mK, so the sampled crossover is at 100 mK. The stored Bose–Einstein occupation of the 6 GHz cavity is $n_{\rm th} = 5.6 \times 10^{-7}$ at 20 mK and 5.95×10^{-2} at 100 mK. The historical hypotheses $n_{\rm th} \approx 1 \times 10^{-4}$ at 20 mK and a fourfold reduction at 100 mK are not reproduced by the explicit-frequency calculation and are not used as manuscript claims.

This result supports a model-specific sampled thermal-retention crossover under the declared bath, preparation, and loss model. It does not establish a universal temperature threshold, a critical temperature, or a platform-independent operating limit, and the occupations are tied to the explicit 6 GHz frequency.

TABLE S23. Sampled peak-QFI retention of the canonical driven row ($\eta = 0.150$, vacuum initial state, $\kappa/\omega_c = 4.1 \times 10^{-2}$, $\omega_c/2\pi = 6$ GHz) relative to its zero-temperature reference. The retention criterion is 80 %; crossover labels are sampled, not critical thresholds.

T (mK)	n_{th}	Peak F_Q	Peak-QFI retention
0	0	1642.1	—
10	3.1×10^{-13}	1642.1	1.0000
20	5.6×10^{-7}	1642.0	1.0000
50	3.2×10^{-3}	1618.0	0.9853
100	6.0×10^{-2}	1273.0	0.7752
150	1.7×10^{-1}	897.2	0.5464
200	3.1×10^{-1}	725.7	0.4420

S24. MULTILEVEL-EMITTER LEAKAGE SCREEN

To test whether the two-level approximation remains adequate under the DCE drive, the producer `scripts/multilevel_emitters.py` evaluates a common oscillator-based emitter interface parameterized by declared transition-frequency ratios for three effective spectra: transmon-like, flux-qubit-like, and fluxonium-like. Each model retains $d_e = 2, \ldots, 5$ emitter levels; static rows use $\eta = 0.15$ and 1.0, and driven rows use drive coefficients $g_d = 0.119\,23$ (canonical) and 0.5 (strong) with $\kappa/\omega_c = 0$ and 4.1×10^{-2} over 40 drive periods, paired with $\eta + \delta\eta$ for a coupling-estimand QFI. The adequacy criterion is declared before scanning: maximum leakage to levels $|2\rangle$ and above below 1 % over the window, recorded as a model choice rather than a universal validity condition. All 76 rows pass the numerical acceptance gate.

No driven $d_e \geq 3$ row meets the criterion. Window-maximum leakage ranges from 2.99 % (fluxonium-like, $d_e = 3$) to 81.5 % (fluxonium-like, $d_e = 5$, strong drive). Table S24 lists the canonical-drive rows at physical loss. Paired two-level versus multilevel comparisons stored in the artifact (38 rows) show the multilevel peak QFI relative change from the two-level baseline ranging from -0.873 (a 0.127× ratio, fluxonium-like) to 7.36 (a 8.36× ratio), with the largest positive deviations at strong drive where leakage is largest; the two-level baseline is therefore not a reliable proxy in either direction.

TABLE S24. Window-maximum leakage of driven multilevel rows at the canonical drive coefficient with physical loss ($\kappa/\omega_c = 4.1 \times 10^{-2}$, 40 drive periods). Effective spectra are declared models, not device simulations; the 1 % criterion is a model choice.

Model	d_e	Max leakage	Adequate at 1 %
Fluxonium-like	3	0.0299	no
Fluxonium-like	4	0.0387	no
Fluxonium-like	5	0.0490	no
Transmon-like	3	0.0775	no
Transmon-like	4	0.0802	no
Transmon-like	5	0.0801	no
Flux-qubit-like	3	0.0873	no
Flux-qubit-like	4	0.1078	no
Flux-qubit-like	5	0.1133	no

This screen quantifies leakage for the explicitly parameterized effective spectra and finite drive window. It does not support a two-level validity guarantee for driven rows at the declared criterion, a definitive platform choice, or a measured flux-noise limit; static rows and two-level baselines cannot leak and provide no driven-adequacy evidence.

S25. EXTENDED HEOM/LINDBLAD AGREEMENT SCREEN

The producer `scripts/heom_extended_validation.py` extends the local bath-model cross-check of Section S22 using the same accepted Spin–Boson format: a Drude–Lorentz bath with the stored coupling convention, a $2\pi J$ golden-rule Lindblad comparator, hierarchy depths $L = 2, \ldots, 5$, and a per-row numerical physicality gate. Rows whose HEOM states fail the trace/Hermiticity/eigenvalue acceptance checks are preserved as `inconclusive_solver_positivity_gate` and are never used for agreement claims. The artifact contains 18 rows, of which 15 are physical and 3 are inconclusive (both tested cutoff extremes and the $T = 0.17$ row). The inconclusive rows document a genuine positivity limitation of the hierarchy solver outside a stable band at low temperature and extreme cutoff; they are recorded rather than repaired.

Among physical rows the hierarchy is converged at $L = 2$: the mean fidelity error is 4.1434×10^{-4} at $L = 2$, and at $L = 3, 4, 5$ it agrees to five significant digits (relative change 4.3×10^{-6}), so the hierarchy is converged at $L = 2$ for the tested rows. At the baseline ($\lambda = 0.02$, $\gamma = 20$, $T = 0.07$, $t_{\mathrm{final}} = 2$), the mean fidelity error is 5.3×10^{-4} and the maximum is 6.6×10^{-4}. Agreement degrades monotonically with coupling (2.6×10^{-3} mean at $\lambda = 0.1$; 7.7×10^{-3} at $\lambda = 0.3$) and remains clean across the tested window. The two driven rows at the DCE frequency ratio are physical and give HEOM-versus-Lindblad QFI relative differences of 1.05 % and 1.09 % at $\eta_d = 0.15$ and 0.10, respectively. The $T = 0.35$ physical row shows a larger mean error (7.7 %), retained as a temperature-scope caveat.

TABLE S25. Extended HEOM/Lindblad screen summary. Errors are Uhlmann-fidelity errors versus the golden-rule Lindblad comparator on the same bath; QFI rows use the coupling estimand.

Family	Rows physical	Mean error range	Max error range	QFI Δ
Hierarchy depth ($L = 2$–5)	4/4	4.14×10^{-4} fixed	6.0×10^{-4} fixed	—
Coupling ($\lambda = 0.02$–0.3)	3/3	5.3×10^{-4}–7.7×10^{-3}	6.6×10^{-4}–9.5×10^{-3}	—
Time window ($t_f = 1$–4)	3/3	4.1×10^{-4}–6.8×10^{-4}	6.0×10^{-4}–1.1×10^{-3}	—
Temperature ($T = 0.07$–0.35)	2/3	5.3×10^{-4}–7.7×10^{-2}	6.6×10^{-4}–1.4×10^{-1}	—
Driven DCE-ratio ($\eta_d = 0.10$, 0.15)	2/2	4.3×10^{-4}–4.8×10^{-4}	5.3×10^{-4}–5.7×10^{-4}	1.09 %, 1.05 %

This screen supports the working Lindblad model for the hierarchy-depth-converged, physical rows and for the two tested driven DCE-ratio rows within the declared errors. It does not generalize beyond the local Spin–Boson/Drude model, the tested rows, or the stated stable parameter band.

S26. STAGED MULTILEVEL DRIVE-AMPLITUDE SWEEP

The original multilevel screen in Section S24 tested only two drive amplitudes ($g_d = 0.119\,23$ and 0.5), both of which failed the 1 % leakage criterion for all driven $d_e \geq 3$ rows. To determine whether a safe operating window exists at lower drive, the producer `scripts/multilevel_drive_sweep.py` evaluates a finer grid of seven drive amplitudes $g_d \in \{0, 0.02, 0.04, 0.06, 0.08, 0.10, 0.119\,23\}$ across three platforms at $d_e = 3$, with $\kappa/\omega_c = 0$ and 4.1×10^{-2}, over 40 drive periods. The artifact `data/multilevel_drive_sweep.json` contains all 42 rows plus a Pareto-frontier analysis.

Table S26 summarizes the maximum g_d that satisfies the 1 % leakage criterion at physical loss for each platform. At zero loss the thresholds are similar (fluxonium-like $g_d \leq 0.06$, flux-qubit-like $g_d \leq 0.04$, transmon-like $g_d \leq 0.02$). The trade-off is monotonic: lower g_d reduces both leakage and QFI enhancement. Nine Pareto-frontier rows span the leakage-QFI boundary.

TABLE S26. Maximum drive coefficient g_d satisfying the 1 % leakage criterion at physical loss ($\kappa/\omega_c = 4.1 \times 10^{-2}$, $d_e = 3$, 40 drive periods). Corresponding QFI peak values are for the coupling estimand.

Platform	Max g_d	QFI at threshold
Fluxonium-like	0.06	85.5
Flux-qubit-like	0.04	77.5
Transmon-like	0.02	54.2

This sweep demonstrates that the two-level approximation has a genuine, platform-dependent safe operating window, but the achievable QFI enhancement at the leakage-acceptable drive amplitude is substantially smaller than at the canonical or strong drive. The 1 % criterion remains a model choice; the models are declared effective spectra, not device simulations.

S27. CAVITY-CHANNEL HEOM VALIDATION

The HEOM cross-checks in Section S22 and Section S25 validate the Lindblad master equation for a *qubit* coupled to a structured Drude–Lorentz bath via σ_x coupling. This does not directly validate the *cavity* single-photon-loss channel $\sqrt{\kappa}\hat{a}$ that produces every headline retention number in the paper. The producer `scripts/cavity_heom_validation_v2.py` addresses this gap: a cavity mode in a Fock space of dimension 12 is coupled through the position quadrature $\hat{x} = \hat{a} + \hat{a}^\dagger$ to a Drude–Lorentz structured bath, and benchmarked against the Markovian Lindblad treatment using $\sqrt{\kappa}\hat{a}$ collapse operators.

The grid spans $\kappa/\omega_c \in \{10^{-4}, 4.1 \times 10^{-2}\}$, Drude cutoffs $\gamma/\omega_c \in \{5, 10, 20, 40\}$, hierarchy depths $L = 2, 3, 4$, temperatures $T = 0.01, 0.05$, and time windows $t_{\text{final}} = 1, 2$ (96 rows). The artifact `data/cavity_heom_validation_v2.json` stores all rows with per-row physicality

diagnostics.

At $\kappa/\omega_c = 10^{-4}$, all 18 physical rows give mean fidelity errors of 9.2×10^{-5}–3.7×10^{-4}, confirming excellent Markovian agreement. The hierarchy shows no meaningful depth dependence at this κ, indicating that non-Markovian cavity-bath memory effects are negligible.

At $\kappa/\omega_c = 4.1\times10^{-2}$, the 18 physical rows give mean fidelity errors of 4.8×10^{-2}–2.4×10^{-1} depending on the Drude cutoff and time window. The hierarchy converges at depth $L = 3$ (relative change $< 1 \times 10^{-5}$ from $L = 3$ to 4). The 60 inconclusive rows fail the positivity gate (minimum eigenvalue down to -1.4), documenting a genuine numerical limitation of the HEOM hierarchy for strong cavity-bath coupling.

The Markovian approximation is therefore approximate at the physical linewidth: in this finite Fock-space, finite-window screen, the physical rows show mean fidelity discrepancies of approximately 4.8×10^{-2}–2.4×10^{-1} across the tested Drude cutoffs and time windows. These discrepancies are a tested model-comparison range, not a propagated statistical uncertainty interval and not a validation of the full Dicke+DCE QFI calculation. The headline retention numbers (16.0% at $N_{\mathrm{cav}} = 6$ and the extension values) should therefore be treated as first-order Markovian estimates whose sensitivity to cavity-bath memory remains model-dependent.

S28. LARGER-N FINITE-SIZE SCALING AND RESOURCE SCREEN

To test whether the superlinear finite-size trend persists under larger N, the producer `scripts/finite_size_resource_scaling.py` evaluates the exact static Dicke ground-state gap-weighted QFI $F_Q = 4\sum_{n>0} |\langle 0|\hat{V}|n\rangle|^2/(E_n - E_0)^2$ with $\hat{V} = (\hat{a}+\hat{a}^\dagger)\hat{J}_x/\sqrt{N}$ at two cavity truncations per size bracket for $N = 4, \ldots, 64$ at $\omega_q/\omega_c = 1.0$ and 1.005, and propagates canonical driven Dicke+DCE rows ($\eta = 1.15$, $g_d/g_0 = 0.15$, $\omega_q/\omega_c = 1.0$, $\omega_d/\omega_c = 2.0$, $N_{\mathrm{cav}} = 6$) with declared pump and emitted-photon proxies. All 52 rows complete; every static row is truncation-converged (relative change typically 1×10^{-11}–1×10^{-14}, all below the declared 5 % criterion), and every driven row passes the numerical physicality gate. The static QFI at the historical anchor $N = 32$, $\omega_q/\omega_c = 1.0$ is 228.6, reproducing the canonical Dicke reference.

Descriptive power-law fits over the declared range $N \in [6, 18]$ give $\alpha = 2.197 \pm 0.039$ ($\omega_q/\omega_c = 1.0$) and $\alpha = 2.216 \pm 0.034$ (1.005), consistent with the historical $\alpha = 2.21$;

dropping the largest in-range point changes α by only 0.004, so the fit is not boundary-dominated within the declared range. Beyond $N \approx 21$ the local growth exponent falls below 1 (0.67 at $N = 24$, 0.34 at $N = 27$, 0.17–0.26 at $N = 30$–36) before recovering to 0.85 at $N = 64$. Single-plateau saturated forms do not describe the data over $N \in [4, 64]$ (maximum relative errors of order 4.6–4.9), so no saturation plateau is reached up to $N = 64$ within this grid. These fits are recorded with uncertainty intervals and boundary-sensitivity entries and are explicitly descriptive.

Table S27 lists the driven resource rows. At a fixed pump proxy $g_d^2 = 0.0298$ (in ω_c units), the lossless peak QFI grows from 785 ($N = 4$) to 1852 ($N = 12$) and 2768 ($N = 24$), while the mean photon number stays near 2–3. Cavity loss degrades the peak and moves it into the window (at $N = 12$: $1852 \to 1073 \to 189$ for $\kappa/\omega_c = 0 \to 1 \times 10^{-2} \to 4.1 \times 10^{-2}$), with the declared emitted-photon proxy $\kappa \int n(t)\, dt$ rising from 0.70 to 2.83 across the same loss rows. A companion producer (`scripts/dicke_eta_sweeps.py`, artifact `data/dicke_eta_sweeps.json`) supplies the static QFI-versus-coupling curves at $N = 4, 8, 16, 32$ shown in the main-text figure: each curve is anchored at $\eta = 1.15$ against the finite-size grid (maximum relative deviation 2.5×10^{-7}), and the sampled maximum shifts with N (≈ 1.40 at $N = 4$, ≈ 1.15 at $N = 16$, ≈ 1.10 at $N = 32$), so the tabulated $\eta = 1.15$ operating point is not the sampled optimum for every N.

TABLE S27. Driven Dicke+DCE resource-accounting rows at fixed pump proxy $g_d^2 = 0.0298$ (ω_c units). The emitted-photon column is the declared proxy $\kappa \int n(t)\, dt$, not measured output; rows are reduced-truncation diagnostics ($N_{\mathrm{cav}} = 6$).

N	κ/ω_c	Peak F_Q	Peak period	Max $\bar{n}$	Emitted proxy
4	0	785.3	10.0	3.25	0
4	1×10^{-2}	469.3	10.0	3.17	0.663
4	4.1×10^{-2}	96.0	6.8	3.06	2.394
12	0	1851.6	10.0	3.19	0
12	1×10^{-2}	1072.6	10.0	2.90	0.696
12	4.1×10^{-2}	188.6	7.7	2.70	2.830
24	0	2768.1	10.0	2.94	0

This screen supports a descriptive superlinear finite-size trend over the declared interval

with uncertainty and boundary sensitivity reported, and characterizes the large-N behavior up to $N = 64$ without a saturation plateau. It does not establish a thermodynamic exponent, a phase transition, a Heisenberg-bound violation, a super-Heisenberg resource, or a restored global bound; the resource accounting is a partial pump/emitted-photon proxy comparison, not a jointly matched operational advantage.

S29. DATA AVAILABILITY

The data and code supporting this study are publicly archived at https://github.com/NanaEngo/Floquet-virtual-state-metrology-dce. The archived package includes the manuscript sources, Supplemental Information, figures, deterministic analysis scripts, environment specification, and JSON data artifacts backing the reported tables and claims. Historical internal audits, generated build files, and local scheduler logs are excluded from the shared package.

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